NFL PLAYING SURFACES PROGRESS REPORT
Ryan Bonin

1. Refined Problem and Innovation Question

My original proposal asked whether the financial/aesthetic benefits of artificial turf outweigh players' health. After refining it further, the investigation now points to a more specific problem: While accounting for the conditions needed to maintain a safe grass field, how can the NFL reduce the avoidable surface-related injury risks? NFL players bear the physical consequences, and their careers are affected while the team and stadium owners manage field performance. My concern remains player health, but as I did prior, I cannot assume owners' motives or conclude that all grass surfaces are safer for players. Going forward, my updated innovation question is: How can stadium operators make the transition towards higher quality natural grass while verifying the safety improvements and addressing the problems that come with the new change? This keeps my original goal intact while shifting from more of an accusation to a question that can lead the way with proper comparisons and allow necessary testing.

2. Work Completed and Evidence Gathered

This is a desk study of three original research studies on NFL injuries. I used the league's policy, the NFLPA report on players, and the stadium operator's report [A1-A6]. I have compared the time periods of the studies, the definition of injuries, the measurement of exposure, and limitations instead of counting the number of articles in favor of grass. The original studies were helpful in assessing the injury claims, whereas institutional sources clarified the policy and feasibility issues. Appendix A has documented the evidence and its few limitations. This is secondary evidence, not primary (zero interviews or conversations with players occurred).

3. Insights, Patterns, and Surprises

One of the patterns that I learned is that studies can reach different conclusions simply because they measure different types of injuries. This was shown as Mack et al. found a 16% higher rate of lower-body injuries per play on artificial turf [A1], while McCormick et al. found no statistically significant difference in season-ending injuries per game [A2]. Venishetty et al. also found higher injury rates and higher odds of season-ending surgery on turf [A3]. Although they all cover injuries, they cover different seasons and are measuring the injuries much differently. That means their results cannot be compared directly. This gives me a reason to investigate grass conversion further without just assuming it would make every field safer, which was a surprise to me as I thought grass was just safer. I learned that players' preferences matter even when the safest playing surface is unknown. Players' strong preference for grass shows that they have concerns that deserve more attention [A5]. Berger's usage of the "Why / What If / How" framework helps me turn those concerns into a practical next step [A7]. This leads me to wonder yet again: Why do players remain concerned when their preferences are just preferences? Another one I wonder is: What if one stadium tested a transition to grass? After this, could its safety, maintenance costs, and event schedule be evaluated and investigated for other teams?

4. Assumptions and Changes in My Thinking

I started a project by assuming turf mainly reflected owners choosing their own wealth over the players' health. Now, that remains a most certainly unverified explanation; I did not use stadium budgets or decision-maker interviews at first, and I also treated grass conversion as a very simple and straightforward

fix for this whole problem. State Farm Stadium's movable field concept demonstrates a workable design [A6], but an already nostalgic and old, existing venue may lack the proper space or infrastructure, causing predicaments. The reality must be investigated rather than inferred just by the abundance of wealth in this football league. The greatest change I noticed is how I evaluate the disagreement. Just because a small result does not prove surfaces are equally safe, and noticing a higher rate for certain injuries does not identify the cause of it directly, since there are overlapping seasons for certain sports and stadiums; this means that studies cannot be counted as fully independent findings [A2-A3]. Therefore, moving forward, keep grass conversion as a serious direction for this project, while abandoning the idea that the same solution can be justified everywhere.

5. Alternatives and Current Direction

Three alternative directions I compared are: immediate league-wide grass conversion, the slow implementation of grass conversion with data, and continued synthetic use with better monitoring. The immediate conversion plan directly addresses player preferences [A5] but lacks a verified operating plan. Monitoring this is also an important comparison option, because the material alone does not relate to performance while leaving player concerns unresolved. The slow implementation idea is strong at this stage, because it makes it testable before applying it to many stadiums and scaling it to the league. This allows independent reviews and player input, and it would track noncontact lower extremity injuries per player, how many days they were out, field performance, and ultimately all player feedback. Next, Berger's "How" stage requires specifying what would change my mind [A7]: inconsistent grass performance or unacceptable operation would trigger a new approach; also, having better sustained outcomes would support expansion. This would be a proposed evaluation, not a completed test. Appendix B compares the alternatives and defines decision gates more thoroughly with data.

6. Remaining Questions, Risks, and Next Steps

Remaining questions and gaps in the ideas I have still are simply the absence of comparable stadium costs, along with uncertainty about injury observations and their differences post-conversion. For the fat portion of this portfolio, I will attempt to seek a grounds manager's explanation of a natural grass surface. He/she plane, crash, recovery, light, drainage, and the effect on other event scheduling at stadiums. I would also go down the route of talking to an athletic trainer or sports medicine researcher to talk about injury severity on each surface, as well as player recovery and confidence after injuries. I will be able to obtain certain operation estimates for this process and develop certain scenarios for each stadium. A single season after this conversion would be way too small of a sample to investigate injury reduction statistically, so, after a few years, a researcher should assess that sample size and investigate before they make any conclusions on this controversial topic. Appendix C specifies evidence needed to decide whether to advance, revise, or reject the idea I am putting out there.

Appendix A Evidence Register

Research reviewed October 16, 2026. Labels A1–A7 link report claims to sources. Facts summarize published material; interpretations and proposed actions are identified separately. No primary fieldwork has been completed for this report.

A1 Higher injury rate in a large NFL cohort

Mack and colleagues analyzed 2012–2016 regular-season injuries using mandated reporting from all 32 teams. Synthetic turf had a 16% higher lower-extremity injury rate per play (IRR 1.16; 95% CI 1.10–1.23).

Source: Mack et al. (2019), American Journal of Sports Medicine, 47(1), 189–196. https://pubmed.ncbi.nlm.nih.gov/30452873/

Limit and project meaning: Observational association; historical fields and seasons cannot establish the risk of every current product. Supports investigating grass conversion, not attributing a particular player's injury to turf.

A2 A different outcome produces a less certain result

McCormick and colleagues found 199 season-ending lower-extremity injuries across 891 NFL games in 2020–2022. Rates were 0.242 per game on artificial turf and 0.199 on grass; the difference was not statistically significant.

Source: McCormick et al. (2024), Translational Sports Medicine, article 6832213. https://pubmed.ncbi.nlm.nih.gov/39553464/

Limit and project meaning: Publicly available records and a narrow outcome. Failure to detect significance does not establish equivalence. Supports separating injury definitions and exposure measures.

A3 Recent research also raises concern

Venishetty and colleagues reported 1.42 lower-extremity injuries per game on artificial turf versus 1.22 on grass in 2021–2022. Their adjusted odds ratio for season-ending surgery was 1.60 (95% CI 1.28–1.99).

Source: Venishetty et al. (2024), Orthopaedic Journal of Sports Medicine, 12(8). https://journals.sagepub.com/doi/10.1177/23259671241265378

Limit and project meaning: Public game reports may miss injuries; practice and preseason injuries were excluded. Odds are not probabilities. The seasons overlap A2, so these are not independent replications.

Appendix A Institutional and Course Evidence

A4 Standards are changing

NFL standards permit grass and synthetic fields, with compliance phased in during 2026–2028.

Source: NFL (December 15, 2025), Safety and Consistency in Field Surfaces.
https://www.nfl.com/playerhealthandsafety/equipment-and-innovation/engineering-technology/safety-and-consistency-in-field-surfaces-nfl-and-nflpa-mandatory-practices

Limit and project meaning: A league policy announcement establishes requirements, not demonstrated injury reductions. Verify implementation and outcomes separately.

A5 Player preference is strong

The NFLPA's 2024 Titans report card states that 92% of surveyed players across the league preferred natural grass to synthetic surfaces.

Source: NFLPA (2024), Tennessee Titans Report Card.
https://nflpa.com/tennessee-titans-report-card-2024

Limit and project meaning: A union-reported preference measure, not a clinical injury rate. The indexed source text was available; full-page access was blocked. Do not describe 92% as the survey response rate.

A6 Grass can coexist with a multipurpose venue

State Farm Stadium describes a natural-grass tray that moves into or out of its retractable-roof venue in about 70 minutes.

Source: State Farm Stadium, About, accessed October 16, 2026.
https://www.statefarmstadium.com/about

Limit and project meaning: Operator documentation establishes a technical example, not the affordability of retrofitting other stadiums or proof of superior injury outcomes.

A7 A course concept guides the investigation

Warren Berger's Why / What If / How framework moves from understanding a problem to exploring alternatives and testing practical action.

Source: Berger, These three questions can help you tackle any problem, author's website.
https://amorebeautifulquestion.com/three-questions-can-help-tackle-problem/

Limit and project meaning: The application to stadium decisions is this project's interpretation. No unverified textbook page numbers or classroom experiences are claimed.

Appendix B Alternatives and Decision Logic

This comparison is analytical work completed for the report. It is not evidence that a stadium approved an option. Costs and injury benefits remain unverified.

Immediate conversion at every venue

Potential value: directly responds to the preference documented in A5. Weak point: no comparable retrofit budgets, recovery plans, or venue-specific performance data. Decision: do not recommend a universal deadline on the present record.

Phased conversion at a feasible venue

Potential value: tests grass quality and operating feasibility before expansion. A6 demonstrates one engineering approach; A1–A3 justify evaluating injury outcomes. Weak point: selection bias, implementation expense, and insufficient injury observations. Decision: prioritize a feasibility study before proposing a live pilot.

Retain synthetic fields with performance monitoring

Potential value: offers a comparison against conversion without assuming a material label determines safety. Weak point: improved test results may not translate into fewer injuries or greater player confidence. Decision: retain as a comparator and possible fallback, contingent on independently assessed outcomes.

Proposed decision gates

1. Feasibility: obtain a site-specific plan and itemized estimates. Revise or reject the venue if adequate grass cannot be sustained through its event calendar.
2. Evaluation: obtain expert review of injury definitions, exposure tracking, sample size, comparison selection, and confounders before testing.
3. Continuation: agree on field-performance and operating limits before launch; investigate failures promptly. Do not interpret zero injuries in a small sample as proof of safety.
4. Expansion: require sustained performance, transparent costs, player input, and enough outcome evidence to justify wider use.

Appendix C Prioritized Learning Plan

The actions below are future work. No outreach, interview, quote, or experiment is represented as already completed.

First resolve operating feasibility

Contact a stadium or professional grounds manager. Ask which light, drainage, recovery, storage, and event-calendar constraints would prevent conversion. Request a sample maintenance schedule and identify the decision-maker who could authorize a change. Output: a venue constraint checklist with documented answers.

Then establish a credible safety comparison

Ask an athletic trainer or injury researcher which injury definitions and denominators are defensible. Request guidance on prior injury, footwear, weather, position, field age, and schedule as confounders. Output: a reviewed study-comparison sheet and proposed evaluation protocol, including whether multiple seasons or venues are needed.

Test the financial assumption

Request separate installation, maintenance, resurfacing, lighting, drainage, and event-disruption estimates. Use the same time horizon for both materials and disclose missing values. Output: a sensitivity analysis; no claimed return on investment without credible inputs. Keep player welfare explicit rather than reducing it to payroll savings.

Understand the user need directly

Seek consent for short conversations with players who have used both materials. Ask about traction, recovery, confidence, and field-condition differences without leading them toward grass. Identify the playing level and avoid generalizing local experiences to NFL players. Output: dated notes separating participant statements from interpretation.

Close the policy and evidence gaps

Verify the policy text and available implementation records, including how field failures are handled. Recheck study corrections and disclosures. Output: an updated source register and a decision explaining whether the pilot should advance, change, or stop.

AI NOTE: I used ChatGPT+ to help with organizing my sources and creating a proper appendix. It also cited the appendix for me in-text. The final ideas, decisions, analysis, and submitted work are my own.